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TASK :- Homograph Detector Tool

What is a Homograph Attack?

A **homograph attack** (also called a **homoglyph attack**) is a type of **phishing or spoofing attack** where **lookalike characters** are used to impersonate **trusted brands, websites, or identities**.

Example:

- Fake domain: g00gle.com
Real domain: google.com
(Uses zero “0” instead of letter “o”)
- Fake domain: paypal.com
(Uses Cyrillic “p”, “a”, “y” — visually identical to Latin “p”, “a”, “y”)

Tool Features:

- User inputs a full sentence or paragraph.
- Tool scans all words in the paragraph.
- It detects words that look like trusted brand names using ASCII homoglyphs.
- It shows:
 - The suspicious word. ◦ Its normalized (decoded) version.
 - The brand or word it's trying to mimic.

Full Python Code: Paragraph-Based Homograph Detector

```
import re

# Trusted brands / keywords to compare against
trusted_words = ['google', 'facebook', 'apple', 'twitter', 'microsoft', 'amazon', 'paypal',
                 'instagram', 'youtube']

# Common ASCII-based homoglyph replacements
homoglyphs = {
```

```

'0': ['o'],
'1': ['l', 'i'],
'3': ['e'],
'5': ['s'],
'7': ['t'],
'@': ['a'],
'$': ['s'],
'!': ['i', 'l'],
'8': ['b'],
'9': ['g']
}

# Normalize suspicious word by replacing homoglyphs
def normalize_word(word):
    normalized = ""
    for char in word.lower():
        if char in homoglyphs:
            normalized += homoglyphs[char][0] # Use first likely substitution
        else:
            normalized += char
    return normalized

# Clean and tokenize input text
def tokenize(text):
    return re.findall(r'\b[\w@!$]+\b', text)

# Main detection function for a paragraph
def detect_homoglyphs_in_paragraph(text):

```

```

words = tokenize(text)
suspicious_results = []

for word in words:
    norm = normalize_word(word)
    for trusted in trusted_words:
        if norm == trusted:
            suspicious_results.append((word, norm, trusted))

return suspicious_results

# CLI runner
def run_detector():
    print("Enter a full paragraph or sentence to scan for ASCII homoglyph attacks:")
    text = input(">> ")
    results = detect_homoglyphs_in_paragraph(text)

    if results:
        print("\nPotential homoglyph attack detected:")
        for original, normalized, match in results:
            print(f"  ➤ '{original}' looks like '{match}' (normalized as: {normalized})")
    else:
        print("No suspicious homoglyph tricks detected.")

# Execute if __name__ ==
"__main__":
    run_detector()

```

output:-

```
🔍 Enter a full paragraph or sentence to scan for ASCII homoglyph attacks:
>> Please don't trust G00gle or Faceb00k links sent on Twltter. Check them twice!

⚠️ Potential homoglyph attack detected:
  ➤ 'G00gle' looks like 'google' (normalized as: google)
  ➤ 'Faceb00k' looks like 'facebook' (normalized as: facebook)

🔍 Enter a full paragraph or sentence to scan for ASCII homoglyph attacks:
>> Siddhi
☒ No suspicious homoglyph tricks detected.
|
```